Statistical Mechanics

Leonard Susskind

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Lecture notes organized by LazyingArt LLC with Video2Book

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Chapter 1

Statistical Mechanics Lecture 1: Probability, Information, and Entropy

These notes follow Leonard Susskind's first lecture on statistical mechanics in the order in which the lecture unfolds. The lecture begins as a manifesto, not as a list of definitions: first the subject is justified, then probability is put on firm ground, then deterministic laws are used to generate probabilities, and only after that do entropy and energy take center stage. The retained lecture frames are kept here as documentary evidence from the original board work; in this edition they are paired with cleaned reconstructions where the board geometry matters, with support from LazyingArt LLC and Video2Book.

1.1 Why Statistical Mechanics?

Susskind begins by refusing to treat statistical mechanics as a secondary subject. It is not merely a bit of nineteenth-century technology bolted onto modern physics. In his telling it is pre-modern, modern, and post-modern all at once. The second law of thermodynamics is announced at once as the guiding light of the whole subject: not yet as something to derive, but as something against which our later reasoning will be tested.

He then asks what statistical mechanics is, but before answering he steps back to the larger question of prediction. The basic laws of physics are laws of evolution. In classical mechanics, and in the predictable aspects of quantum mechanics as well, if we know the initial condition of a closed system and if we know the laws of motion, then in principle prediction is complete. That is the ideal of exact microscopic description.

The difficulty is that exact microscopic prediction can become practically useless long before it ceases to be exact. A complete list of the positions and velocities of every molecule in a room would be too long to grasp and too unstable to handle. So there is an immediate gap between exact law and useful knowledge. Statistical mechanics enters precisely in that gap. It is what we use when exact initial data are unavailable, when the system is not really closed, or when the detail is so overwhelming that a coarser description is the only sensible one.

This is why Susskind immediately turns to the example of a gas in a box. There are things we can predict very well, such as pressure and average energy, and there are things we cannot predict in their exact occurrence, such as the precise time at which a large fluctuation will appear. We can

predict the probability of the fluctuation, but not the moment at which it occurs. The coin toss is the same story in miniature: we may be quite sure that roughly half of a billion tosses will be heads, and yet we cannot say when a long streak of heads will begin.

The lecture also pauses over a more cultural point. The great physicists were masters of statistical mechanics, not only because it was useful, but because it is beautiful. That matters here. The subject is not being introduced as emergency bookkeeping for ignorance; it is being introduced as a mathematically serious way of thinking about physical systems when ignorance itself has structure.

1.2 Probability As The Starting Language

Only after this opening does Susskind begin what he calls a mathematical interlude. In this lecture it is really the starting point. We imagine a space of possibilities, and we label its elements by a discrete index i. The i-th possibility may be an outcome of an experiment, or it may be the exact microscopic state of a system. The point is that at a given time the system is in one precise state, but we do not know which one.

To each state we assign a probability $p(i)$. The first rules are the obvious ones,

$$p(i) \geq 0, \tag{1.1}$$

$$\sum_i p(i) = 1. \tag{1.2}$$

These states are taken to be mutually exclusive. If one state occurs, the others do not. That assumption is simple, but the lecture later returns to it explicitly in the question period, so it is worth stating clearly at the start.

Susskind then ties probability to repeated trials. If $N(i)$ is the number of times the i-th outcome occurs among N_{trials} repetitions, then the law-of-large-numbers interpretation is

$$p(i) = \lim_{N_{\text{trials}} \to \infty} \frac{N(i)}{N_{\text{trials}}}. \tag{1.3}$$

He is careful here: this is a physical hypothesis, not a logical necessity. We never perform infinitely many trials. We idealize. But in large systems and repeated experiments the idealization is good enough that it becomes one of the central working assumptions of physics.

Now attach to each state a quantity $f(i)$. It may be an energy, a momentum, or simply a number we invent in order to discuss an average. Susskind uses bracket notation for the average,

$$\langle f \rangle = \sum_i f(i)\, p(i). \tag{1.4}$$

In frequency language the same quantity can be written as

$$\langle f \rangle = \lim_{N_{\text{trials}} \to \infty} \frac{1}{N_{\text{trials}}} \sum_i f(i)\, N(i). \tag{1.5}$$

If we momentarily use the board's letter F instead of the spoken f, this is the same structure that remains visible on the left side of the later lecture frame:

$$\langle F \rangle = \sum_i F(i)\, p(i) = \lim_{N_{\text{trials}} \to \infty} \frac{1}{N_{\text{trials}}} \sum_i F(i)\, N(i). \tag{1.6}$$

The cropped board retains precisely the pieces $F(i)\,p(i)$ and $F(i)N(i)/N$, which is enough to identify the idea.

The coin example makes the definition concrete. If

$$f(\text{heads}) = +1, \qquad f(\text{tails}) = -1, \tag{1.7}$$

then for a fair coin,

$$\langle f \rangle = (+1)\frac{1}{2} + (-1)\frac{1}{2} = 0. \tag{1.8}$$

This is a small point, but an important one: the average value need not itself be a possible outcome of the experiment. Zero is not one of the faces of the coin toss. It is the weighted average of the distribution.

That completes the leveling of the ground. The lecture has not yet done statistical mechanics proper. It has fixed the language in which the rest of the subject will be spoken: states, probabilities, frequencies, and expectation values.

1.3 Symmetry, Experiment, And Dynamical Probability

At this point Susskind restarts with coins. Why do we assign probability 1/2 to heads and 1/2 to tails? In the first instance, because of symmetry. Heads and tails are related by a symmetry of the coin, and apart from tiny imperfections there is no reason for one to occur more often than the other. The same thought applies to a symmetric die.

But symmetry is not always available. If the die is misshapen, or if the system under discussion has no obvious invariance, then symmetry no longer dictates the answer. In that case the lecture's next answer is experiment: throw the die again and again, count the outcomes, and use the measured frequencies.

Then comes the crucial shift. There is another possible origin of probability, and it comes from deterministic evolution itself. We may know a law of motion without knowing the starting point. That is enough to produce a probability distribution.

Suppose the system has six states labeled by colors, and suppose the update rule is

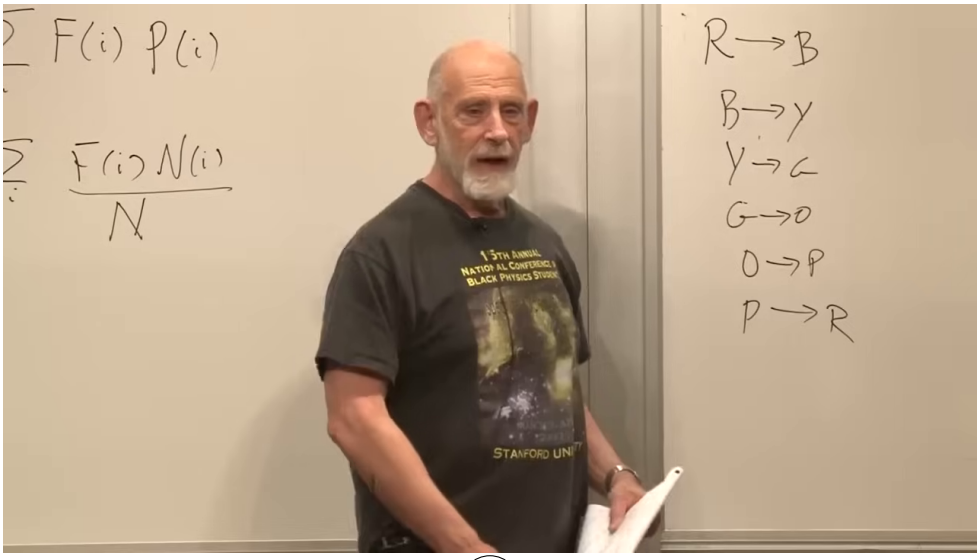
$$\begin{aligned} &R \to B, && B \to Y, && Y \to G, \\ &G \to O, && O \to P, && P \to R. \end{aligned} \tag{1.9}$$

Here P stands for purple. The object carrying these labels need not be symmetric at all. The lecture is explicit about this: what matters is not the symmetry of the die, but the existence of a definite update rule.

Assume that each update takes the same discrete time interval. Then if we sample the system at an unknown instant, without knowing where on the cycle it began, the probabilities are

$$p(R) = p(B) = p(Y) = p(G) = p(O) = p(P) = \frac{1}{6}. \tag{1.10}$$

The system spends one sixth of its time in each state, and that is the whole reason for the uniform distribution.



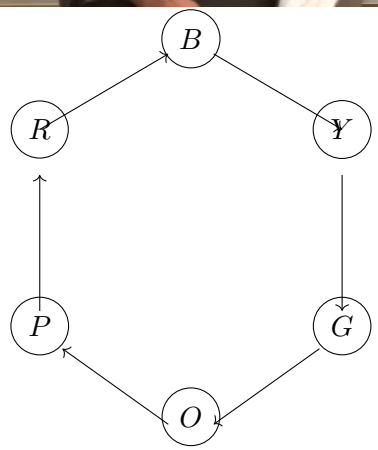


Figure 1.1: At the moment Susskind says "I'm going to draw this law as a diagram," the board still shows earlier probability notation on the left and the six-step update list on the right. Below it we redraw the same law as a closed cycle.

The lecture then strengthens the point by changing the order of the cycle while keeping the property that each color is visited once per period. The conclusion remains the same. The probability 1/6 does not depend on knowing the microscopic starting point, and it does not even depend on the detailed ordering of the cycle. It depends only on the orbit structure: one closed cycle, equal time per step, one visit per state.

1.3.1 Question & Answer

Question. If the system is not symmetric, why can each state still have probability 1/6?

Answer. Because the probability here is not being assigned by the shape of the object. It is being assigned by the time spent in each state. The lecture is very explicit on this point. If the system moves deterministically through a single six-step cycle, and if we are ignorant of the starting moment, then a random flash snapshot has to weight the six time-slots equally. The relevant symmetry is

therefore a symmetry of the orbit in time, not a symmetry of the die in space.

1.4 Cycles, Conserved Quantities, And Two Connected Systems

The next move complicates the state space in exactly the right way. The six states need not lie on one big cycle. They may split into disconnected cycles. Suppose, for example, that one law produces the three-cycle

$$R \to B \to G \to R \tag{1.11}$$

and another disconnected three-cycle

$$P \to Y \to O \to P. \tag{1.12}$$

If we know the system is on the first cycle, then

$$p(R) = p(B) = p(G) = \frac{1}{3}, \qquad p(P) = p(Y) = p(O) = 0. \tag{1.13}$$

If we know it is on the second cycle, the reverse is true.

But now a new issue appears. Dynamics alone does not tell us the probability of being on the first cycle rather than the second. That extra information has to come from elsewhere. Susskind's way of saying this is that one must supply some other probability, some prior information that chooses between the cycles. In the lecture the cycles are labeled $+1$ and -1, and that label is interpreted as a conserved quantity. It could be a toy quantity, but it could just as well be energy.

So let p_+ be the prior probability for the $+1$ cycle and p_- for the -1 cycle, with

$$p_+ + p_- = 1. \tag{1.14}$$

Then the probabilities inside each sector are still uniform, but the overall state probabilities are

$$p(R) = p(B) = p(G) = \frac{p_+}{3}, \qquad p(P) = p(Y) = p(O) = \frac{p_-}{3}. \tag{1.15}$$

This is the first real appearance of a conservation law. A conserved quantity partitions the space of states into sectors, and deterministic motion remains trapped inside one sector. Once that is so, there are really two distinct questions. First: within a given sector, how is time distributed among the available states? Second: what is the prior probability of being in that sector at all? The first question is answered by cycling. The second is not.

The lecture even remarks that the cycles need not have the same size. One sector might contain four states while another contains two. That changes the probabilities within a sector, but it does not change the logic. A conserved quantity partitions the configuration space, and statistical mechanics must somehow determine the relative weights of the sectors.

This is the bridge to honest energy. A single closed box may be treated as a universe unto itself, with a conserved energy. Two isolated boxes have two separately conserved energies. But if the boxes are connected by a narrow passage so that energy can pass between them, then only the total energy is conserved, and the new statistical question is forced upon us: given the total energy, what is the probability that one subsystem has a specified share?

If the two boxes are equal, one expects equal energy on average. But that is only the first moment of the problem. The real question is the distribution itself. This is exactly where the lecture wants us: dynamics has produced sectors, conservation laws have appeared, and now the statistical question is how to assign probabilities to those sectors in a physically meaningful way.

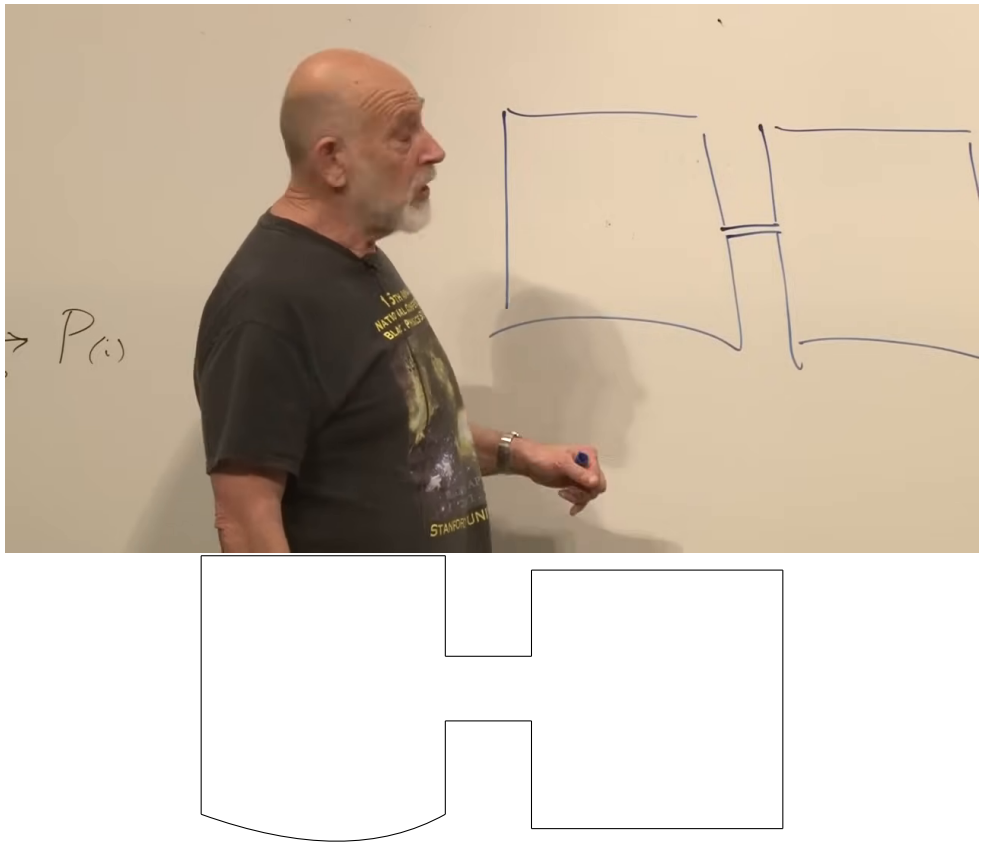


Figure 1.2: The lecture's sketch of two connected subsystems. Once the two parts can exchange energy, the statistical problem becomes the problem of how a fixed total energy is distributed between them.

1.5 Good Laws, Bad Laws, And The Minus First Law

At this point Susskind takes a step that is diagnostic rather than constructive. Before deciding how to assign probabilities, he asks what kinds of laws are even allowed. Some laws are bad, not in a moral sense, but in the sense that physics forbids them.

The prototype is the rule that sends every state to the same state:

$$\begin{aligned} &R \to R, && B \to R, && Y \to R, \\ &G \to R, && O \to R, && P \to R. \end{aligned} \tag{1.16}$$

This law predicts the future in a trivial way. Wherever you are, next you are at red. But it destroys the past. If the present state is red, you can no longer reconstruct where you came from. Distinctions between states have been erased.

That is what Susskind means by loss of information. He gives the principle a name of his own: the minus first law of physics. Information is never lost. Differences between states propagate with time. A good law is therefore one in which every state has one incoming arrow and one outgoing arrow. In discrete language that is the mark of reversibility: one arrow tells you where you are going, the other tells you where you came from.

The lecture then reformulates the same point probabilistically. Suppose that at some moment M out of N states carry equal nonzero probability $1/M$, while the remaining states have probability zero. Under a good law, the occupied states may reshuffle, but there will later still be exactly M occupied states, each with probability $1/M$. Under a bad law such as (1.16), that number can collapse. Distinct possibilities are merged. This observation is the bridge from information conservation to entropy.

In continuum mechanics the corresponding theorem is Liouville's theorem. A region of occupied phase space may stretch, shear, and fold, but its phase-space volume is preserved. More than that, a probability distribution that is initially uniform on the occupied region remains uniform on the evolved region. The discrete rule "M occupied states stay M occupied states" and the continuous rule "occupied phase-space volume is preserved" are the same principle in two different languages.

1.5.1 Question & Answer

Question. Why does friction not really violate conservation of information?

Answer. Because friction only looks like a fundamental information-destroying law when we describe too few degrees of freedom. Susskind's schematic example is a viscous-drag equation of the form

$$m\ddot{x} = -\gamma\dot{x}. \tag{1.17}$$

Taken literally as a fundamental law for a closed many-particle system, such an equation would be disastrous: every particle would drift toward rest, energy would disappear, and the system would evolve toward a simpler, less informative description. It would be the continuum analog of "everything goes to red."

But real friction is not that. When an eraser slows down on a table, the lost macroscopic motion is recorded in microscopic variables of the eraser, the table, and the surrounding air. If we look only at the visible coordinate x and its conjugate momentum, the distribution seems to contract. In the full higher-dimensional phase space, however, the distribution spreads into hidden directions. The apparent collapse is a projection. Liouville volume is preserved in the full closed system even though it is not preserved in the macroscopic variables we happen to be watching.

That is the point of the example. The lecture does not need the exact drag coefficient. It needs the conceptual lesson: apparent irreversibility at the coarse level is compatible with information conservation at the microscopic level.

1.6 The First Law And The Additivity Caveat

After the minus first law, Susskind deliberately skips the zeroth law and states the first law in its simplest possible form. Energy is conserved:

$$\frac{dE}{dt} = 0. \tag{1.18}$$

That is all the first law says for a closed system.

If the system is built from two weakly interacting parts, then the total energy is approximately additive,

$$E \approx E_1 + E_2, \tag{1.19}$$

and differentiation gives

$$\frac{dE_1}{dt} + \frac{dE_2}{dt} = 0, \tag{1.20}$$

$$\frac{dE_1}{dt} = -\frac{dE_2}{dt}. \tag{1.21}$$

Susskind explicitly writes the second form to make the physical interpretation vivid: what leaves one subsystem enters the other.

He is equally explicit about the caveat. Exact additivity is not always true. If two bodies interact gravitationally, for example, the total energy is not merely the kinetic energy of one plus the kinetic energy of the other. There is also the potential energy of interaction, and it belongs to the pair rather than to either body separately.

Why then is additivity so useful in thermodynamics? Because in many large systems the interaction term is a surface effect while the bulk energy is a volume effect. If we divide a large body into blocks, the energy in a block scales with its volume, whereas the interaction between neighboring blocks scales with the contact area of their surfaces. For large enough blocks the interaction term is negligible by comparison, and the additive approximation becomes excellent. The first law is always true. The two-subsystem form is true when interaction energies are small enough to ignore.

This short section matters because it fixes the hierarchy of the lecture. Energy conservation is fundamental, but in this first lecture it is being inserted only after information conservation has been clarified. Entropy is then allowed to return in a sharper form.

1.7 Entropy As A Measure Of Ignorance

1.7.1 Occupied States And The First Definition

Now the lecture explicitly returns to entropy. Up to this point Susskind has defined it only in a special case, but he has already motivated the key ingredient: the number of states that are genuinely available to us given our incomplete knowledge.

Suppose there are N possible states altogether, but only M of them have nonzero probability, and suppose those M occupied states are equally likely. Then

$$p_i = \frac{1}{M} \quad \text{for the occupied states,} \qquad p_i = 0 \quad \text{for the others.} \tag{1.22}$$

The entropy is defined by

$$S = \log M. \tag{1.23}$$

This is already enough to capture the lecture's main idea. If $M = N$, then all states are equally probable and our ignorance is maximal:

$$S_{\max} = \log N. \tag{1.24}$$

If $M = 1$, then the system is known to be in one specific state and the entropy is zero.

Susskind keeps returning to the same interpretation: M measures ignorance. The bigger M is, the less we know. And because a good law only reshuffles occupied states without changing their number, M is conserved under exact microscopic evolution. In that sense entropy is conserved if we can follow the system perfectly.

But we usually cannot follow the system perfectly. We lose track of phases, timing, microscopic positions, and details of the dynamics. Then the entropy increases, not because the equations have destroyed information, but because our description has become coarser. That is why Susskind insists that entropy enters before temperature and, in a certain sense, before energy in the pedagogical order of the lecture. It is the measure of our statistical ignorance.

1.7.2 General Probability Distributions

The lecture then says, in effect: we are not finished with entropy. The special rectangular distribution is too narrow a case. Most probability distributions are not flat on their occupied set. To show what a more general situation looks like, Susskind draws the states schematically along a horizontal axis and plots the corresponding probabilities on the vertical axis.

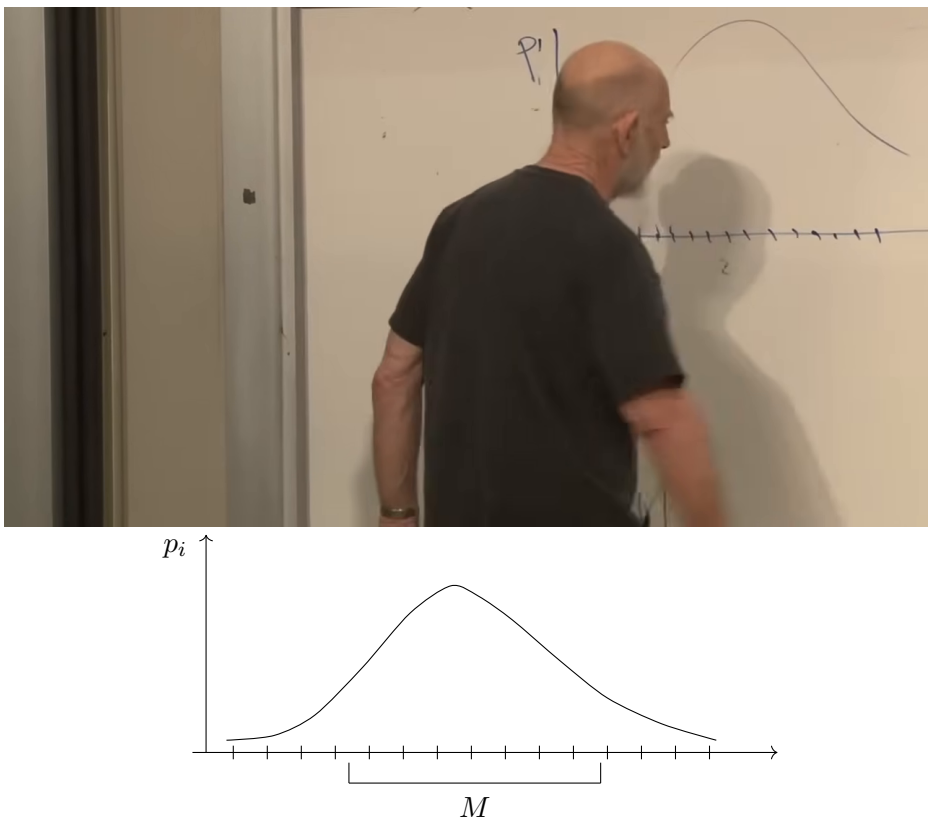


Figure 1.3: The lecture's schematic probability graph over discrete state labels. The picture is qualitative, but its purpose is precise: a narrower distribution represents fewer significantly occupied states and therefore less entropy, while a broader one represents more and therefore more entropy.

The definition that handles the general case is

$$S = -\sum_i p_i \log p_i. \tag{1.25}$$

Susskind immediately remarks that this is minus the average of $\log p_i$. That is why the earlier expectation-value machinery mattered. The new entropy formula is built directly out of it.

The minus sign is not mysterious. Probabilities are numbers less than 1, so their logarithms are negative. The overall minus sign is there so that entropy comes out positive in the cases of interest. The lecture then checks the formula against the original $S = \log M$.

If $p_i = 1/M$ on the occupied states and $p_i = 0$ elsewhere, then the states of zero probability contribute nothing because

$$\lim_{p \to 0^+} p \log p = 0. \tag{1.26}$$

For the occupied states,

$$\begin{aligned} S &= -\sum_{\text{occupied } i} \frac{1}{M} \log\left(\frac{1}{M}\right) && (1.27) \\ &= -M \left[\frac{1}{M} \log\left(\frac{1}{M}\right)\right] && (1.28) \\ &= -\log\left(\frac{1}{M}\right) && (1.29) \\ &= \log M. && (1.30) \end{aligned}$$

So the new definition reduces exactly to the old one in the special case where it must.

This is the moment at which the lecture gives its qualitative interpretation of the graph. The narrower the distribution, the smaller the entropy. The broader the distribution, the larger the entropy. The later frame below simply reinforces that visual point by drawing attention to the width marker M.

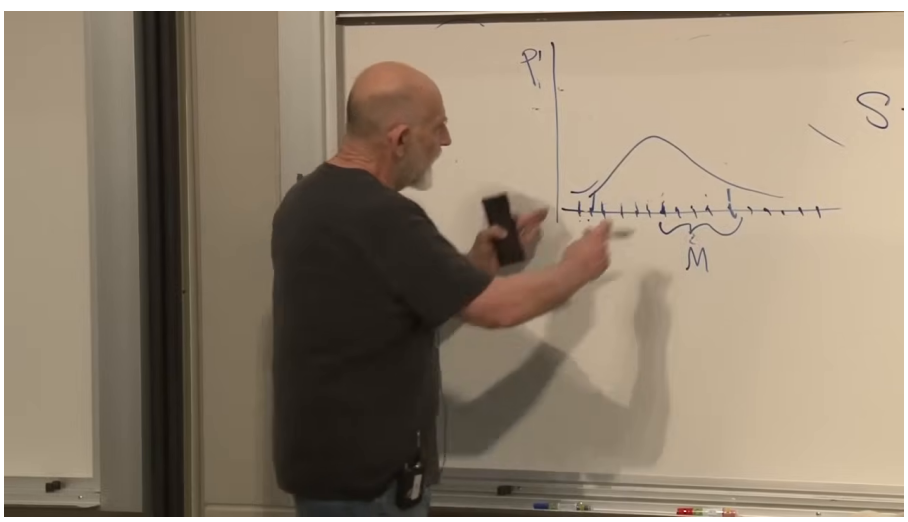


Figure 1.4: A later frame of the same graph. The lecturer's gesture singles out the spread of the distribution, which is exactly the feature the surrounding discussion interprets as controlling the entropy.

1.7.3 Coins As Explicit Examples

The lecture now returns once more to coins, because they let us see the definitions work in explicit examples. To keep the notation straight, let n denote the number of coins, and let N denote the total number of states of the n-coin system. Then

$$N = 2^n. \tag{1.31}$$

If we know nothing at all, then every one of the 2^n states is equally probable. Therefore

$$S = \log(2^n) = n \log 2. \tag{1.32}$$

This is the maximal entropy of the n-coin system.

If the microscopic state is known completely, then only one state has nonzero probability. In that case

$$S = 0. \tag{1.33}$$

Perfect knowledge means zero entropy.

The more interesting example is the one Susskind works through with the class. Suppose all n coins are heads except for exactly one tail, but we do not know which coin is tails. Then there are exactly n states with nonzero probability, all equally weighted, and therefore

$$S = \log n. \tag{1.34}$$

This example matters because it shows that with natural logarithms entropy is not, in general, an integer multiple of $\log 2$.

The lecture also addresses the base of the logarithm. In these notes, as in the lecture, log means the natural logarithm. In information theory one often uses base 2, in which case the entropy of a completely unknown single coin is 1 bit. The choice of base changes only an overall numerical factor. It does not change the structure of the theory.

1.7.4 Continuous Phase Space

The lecture then enlarges the whole discussion from discrete state spaces to classical phase space. A microscopic state is now a point specified by coordinates and momenta. Susskind uses both x and q verbally for coordinates; we will simply write the state space abstractly as phase space Γ, and we will denote the phase-space volume element by $d\Gamma$.

Suppose first that the probability distribution is uniform on some occupied blob of phase space and zero outside. Then the continuous analog of counting occupied states is measuring the phase-space volume of the occupied region. The entropy is defined by

$$S = \log V_\Gamma, \tag{1.35}$$

where V_Γ is the phase-space volume of the blob.

For a general probability density P on phase space, the normalization condition is

$$\int d\Gamma \, P = 1, \tag{1.36}$$

and the entropy becomes

$$S = -\int d\Gamma \, P \log P. \tag{1.37}$$

This is exactly the continuous analog of the discrete formula. One simply replaces the sum over discrete states by an integral over phase space.

Susskind's interpretation remains the same. In first conceptual approximation, the entropy measures the logarithm of the volume of the probability blob. More generally, it measures how spread out the probability density is in phase space. It is still a property of a probability distribution, not a property of a microscopic state by itself. A single exact microstate has no entropy except in the trivial sense that the entropy of the corresponding probability distribution is zero.

1.7.5 Correlation And Additivity

The lecture closes by sharpening the interpretation of entropy through correlation. If we truly know nothing about a collection of coins, then measuring one coin tells us nothing about the others. The remaining probabilities are unchanged. That is the uncorrelated case.

But if we know that exactly one coin among n is tails, then a measurement of one coin changes our knowledge of the rest. If the measured coin is tails, the others are certainly heads. If the measured coin is heads, then the probability that any given remaining coin is the tail becomes

$$p(\text{tail on a given remaining coin} \mid \text{measured head}) = \frac{1}{n-1}. \tag{1.38}$$

That is Susskind's definition of correlation: measuring one thing changes the probability distribution for another thing.

This links directly back to the earlier discussion of conserved sectors. Once we determine a conserved quantity, the admissible set of states changes, and so the probability distribution changes with it. Correlation is therefore not an ornamental afterthought. It is part of the central logic of statistical mechanics: knowledge about one part of a system can constrain what remains possible elsewhere.

Entropy is additive when there is no correlation. For uncorrelated subsystems the total entropy is the sum of the entropies of the parts. When there are correlations, additivity fails in exactly the way one would expect: some of the ignorance in one subsystem is no longer independent of the ignorance in another.

The audience then asks about the relation to information theory. Susskind's answer is that the Boltzmann and Shannon entropies are the same formula up to the base of the logarithm and a sign convention. What is called entropy here is therefore a measure of lack of information. It is not the Heisenberg uncertainty principle. The lecture is explicit on that point: the entropy under discussion concerns probabilistic ignorance encoded in a mixed distribution over states, not the distinct quantum uncertainty attached to a pure state.

1.8 Summary

The first lecture unfolds in a very deliberate order. Statistical mechanics is first justified as a deep and central subject. Then probability is reviewed as the language of incomplete knowledge. Then deterministic laws are shown to generate probabilities through ignorance of the starting point. Then

conservation laws partition the state space, and the conservation of information is elevated to the minus first law. Only after that does the lecture state the first law and return to entropy.

Entropy begins as

$$S = \log M, \tag{1.39}$$

the logarithm of the number of equally probable occupied states. It then broadens to

$$S = -\sum_i p_i \log p_i, \tag{1.40}$$

and finally to the continuous phase-space form

$$S = -\int d\Gamma \, P \log P. \tag{1.41}$$

In every case the meaning is the same: entropy is not a bare property of a microscopic state; it is a measure of our probabilistic ignorance, organized over the space of possible states.

That is the real achievement of the lecture. By the end, entropy has been made concrete enough to calculate, broad enough to survive the passage from discrete to continuous systems, and precise enough to prepare the next step, where temperature and the Boltzmann distribution will enter.

Chapter 2

Temperature as Energy and as the Derivative of Entropy

This lecture unfolds in a deliberately careful order. We do not begin by defining temperature. We begin by removing the human conventions that make temperature look more mysterious than it is. Once temperature has been recast as an energy, and entropy as a dimensionless quantity, the statistical-mechanical definition can appear in its proper form. These notes follow Leonard Susskind's Lecture 2 closely, while preserving selected board images as documentary evidence; those visual materials are retained here with the help of LazyingArt LLC and Video2Book.

2.1 Units, Conversion Factors, and Why Temperature Is Really Energy

The lecture opens with a question that sounds elementary but is not: what is Boltzmann's constant? Susskind's answer is that k_B is, first of all, a conversion factor. In that respect it belongs to the same family as the speed of light. One may use c to convert distances into times; one uses k_B to convert human temperature units into energy units.

This is worth emphasizing because the ordinary temperature scales are historical constructions. Kelvin is measured from absolute zero, Celsius from the freezing and boiling points of water, and the size of a degree was chosen for convenience. Those units are perfectly useful, but they are not the units that microscopic physics would have invented from scratch.

The natural unit of temperature is energy. The lecture insists on this before anything else happens.

A dilute gas gives the cleanest example. The average kinetic energy of one molecule is

$$\langle E_{\text{kin}} \rangle = \frac{3}{2} k_B T_{\text{Kelvin}}. \tag{2.1}$$

The factor of 3 comes from the three directions of space; the factor of 1/2 is the familiar one from kinetic energy. What matters here is that the combination $k_B T_{\text{Kelvin}}$ has units of energy.

Susskind then pushes the point with a slightly comic but very effective example. Put a bowling ball into a fluid in thermal equilibrium. Its speed will be tiny compared with that of a molecule, but its characteristic kinetic energy is set by the same temperature. Equal temperature does not mean equal velocity. It means equal energy scale.

For that reason we define a more fundamental temperature variable by

$$T := k_B T_{\text{Kelvin}}. \tag{2.2}$$

Now temperature is itself measured in energy units. The same kinetic-energy relation becomes

$$\langle E_{\text{kin}} \rangle = \frac{3}{2} T. \tag{2.3}$$

This is the first payoff of the lecture. Once temperature is written in fundamental units, k_B disappears from the formulas. It has not gone away physically. It has been absorbed into the choice of units.

There is also an important conceptual side remark here. The numerical value of k_B is not an absolute thing independent of convention. If one changed from Kelvin to some other temperature scale, the numerical value of k_B would change. That is precisely what conversion factors do.

2.2 Entropy Units and the Disappearance of k_B

Susskind then performs the same cleanup for entropy. Historically, entropy first appeared in thermodynamics in engineering units. Carnot did not know anything about microscopic states or information. He had a macroscopic quantity, useful for steam engines, with units of energy divided by temperature. If temperature is measured in Kelvin, that means joules per Kelvin.

Statistical mechanics uses a different normalization. For a discrete probability distribution $\{p_i\}$, the entropy is

$$S = -\sum_i p_i \log p_i. \tag{2.4}$$

This quantity is dimensionless. It is measured in the same sense that information is measured: one bit corresponds to $\log 2$.

The lecture therefore distinguishes two entropies: Carnot's engineering entropy and Boltzmann's statistical entropy. Their relation is

$$S = \frac{1}{k_B} S_{\text{Carnot}}. \tag{2.5}$$

Equivalently,

$$S_{\text{Carnot}} = k_B S. \tag{2.6}$$

So the same constant k_B converts not only ordinary temperature into energy, but also engineering entropy into dimensionless entropy. Once both conversions have been made, the later formulas can be written in their cleanest form. This is why the long opening discussion of units is not a detachable preface. It prepares the ground for the relation

$$dE = T\, dS$$

so that when it finally appears it does not look like an arbitrary formal trick with a missing constant.

Remark 2.1. From this point on we write T for temperature in energy units and S for dimensionless statistical entropy. If one ever wants to go back to human units, one restores $T = k_B T_{\text{Kelvin}}$ and $S_{\text{Carnot}} = k_B S$.

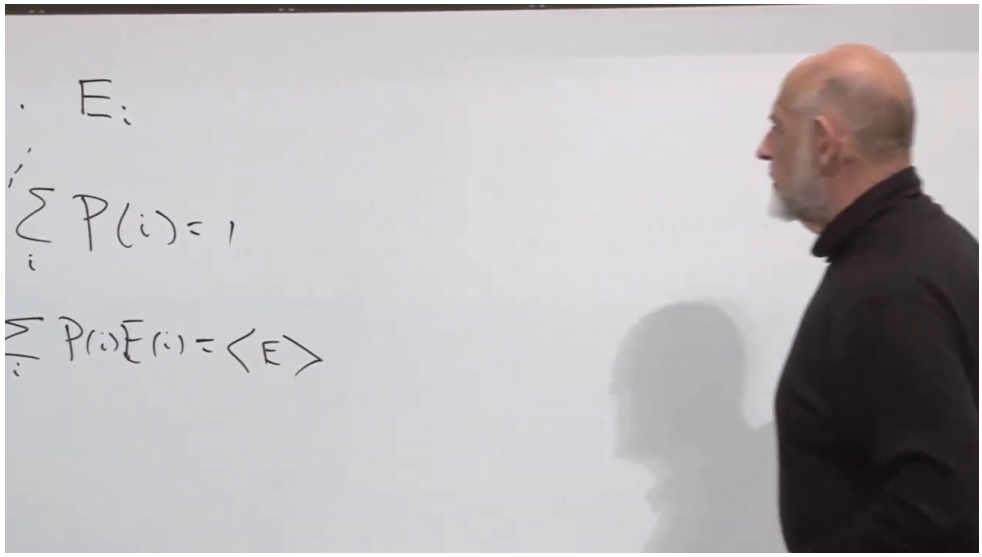


Figure 2.1: Normalization and average energy. The board mixes E_i, $E(i)$, and $\langle E \rangle$, but the content is the same: probabilities sum to one, and the weighted sum of state energies gives the average energy.

2.3 Thermal Equilibrium as a Probability Distribution Over States

Having cleared away the units, Susskind makes the sharp pivot: thus far he still has not told us what temperature is.

The next step is thermal equilibrium. Here the system should not be imagined as forever frozen in one exact microstate. Quite the opposite: the system may be exchanging energy with an environment, or heat bath, and its instantaneous energy may fluctuate. Thermal equilibrium means that after enough time the statistical description settles down. On the average, energy is flowing neither way.

Let the microscopic states of the system be labeled by i, with energies E_i. In equilibrium there is a probability $P(i)$ that the system is in the i-th state. Those probabilities satisfy the normalization condition

$$\sum_i P(i) = 1, \tag{2.7}$$

and the average energy is

$$\sum_i P(i)E_i = E. \tag{2.8}$$

On the board the same equation is written in a mixed notation as $\sum_i P(i)E(i) = \langle E \rangle$. In these notes we standardize to E_i for the energy of the i-th state and E for the average energy.

The word *average* matters. The energy is not assumed to be sharply fixed at every instant. It fluctuates because the system exchanges energy with its surroundings. If one measures the system repeatedly, one gets a distribution of outcomes. The equilibrium quantity is the average of that fluctuating energy.

This is the first genuinely statistical description of equilibrium in the lecture. We are no longer talking about a thermometer reading in the abstract. We are talking about a probability distribution over microstates.

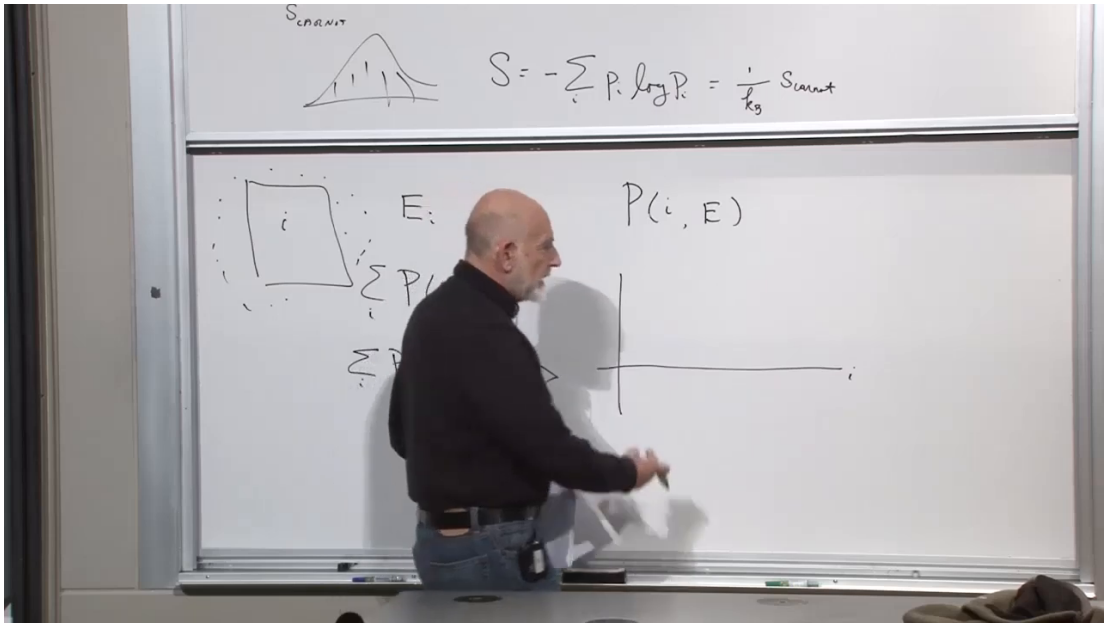


Figure 2.2: Setting up the probability distribution $P(i, E)$. The board is in transition: earlier entropy and average-energy formulas remain on the left while the new graph appears on the right.

2.4 A One-Parameter Family of Equilibrium Distributions

Once the equilibrium probabilities have been introduced, the lecture asks the next question: what does the probability distribution depend on? The answer is that it depends, above all, on the average energy.

This is a physical statement before it is a formal one. Heat a pot of water on the stove, and you change its average energy. Cool it in the snow, and you change the average energy in the opposite direction. The equilibrium distribution changes with that parameter. So equilibrium is not one distribution but a family of them.

We therefore write

$$P(i, E), \tag{2.9}$$

where the symbol E now labels the average energy of the distribution, not the energy E_i of an individual microscopic state.

For each member of the family, the defining relations are

$$\sum_i P(i, E) = 1, \tag{2.10}$$

$$\sum_i P(i, E)\, E_i = E. \tag{2.11}$$

These are the cleaned-up equations that correspond to the board discussion. The board still shows the older unparameterized formulas on the left, but the spoken argument has already moved to the one-parameter family.

The lecture then shifts from formulas to pictures. On the right-hand side of the board Susskind draws axes labeled $P(i, E)$. At first the horizontal variable may be thought of simply as the state

Figure 2.3: A restrained reconstruction of the first graph. At this stage the horizontal axis may be read either as the state label i or, after ordering the states by energy, as a proxy for E_i.

label i. But once the states are ordered by increasing energy, the same axis may also be read as tracking E_i.

The important point is that equilibrium depends on a parameter. For each value of the average energy there is a different equilibrium distribution. The lecture does not derive that family yet; it takes it as a matter of experience and postpones the derivation of the Boltzmann distribution.

2.4.1 Question & Answer

Question. Isn't this circular? The average energy is computed from the probability distribution, but the probability distribution is now being labeled by the average energy.

Answer. Susskind's answer is cautious and deliberately provisional. For the moment we assume that there is a one-parameter family of equilibrium distributions. Each distribution has an average energy. If no two members of the family have the same average energy, then the average energy can be used to label the family. The notation $P(i, E)$ is therefore a convenient naming convention for a family that is being assumed, not yet derived.

Several classroom clarifications sharpen the point. First, Susskind says *average* energy rather than *total* energy because the system is imagined to be in contact with a heat bath, so its energy fluctuates. Second, to make the graph readable, he asks us to adopt a simple working assumption: label the states in increasing order of energy, and ignore exact degeneracies. That is why the horizontal axis may slide from a state label i to something like E_i without changing the logic of the picture.

2.5 Entropy as a Function of Average Energy

At this point the lecture inserts what Susskind calls "one more thing." Each member of the family $P(i, E)$ has its own entropy. Therefore entropy becomes a function of the average energy:

$$S(E) = -\sum_i P(i, E) \log P(i, E). \tag{2.12}$$

This is not yet a universal function of energy. It depends on the particular family of equilibrium distributions for the system under discussion. But once the family has been granted, entropy is no longer an independent object floating beside it. It becomes a function built from that family.

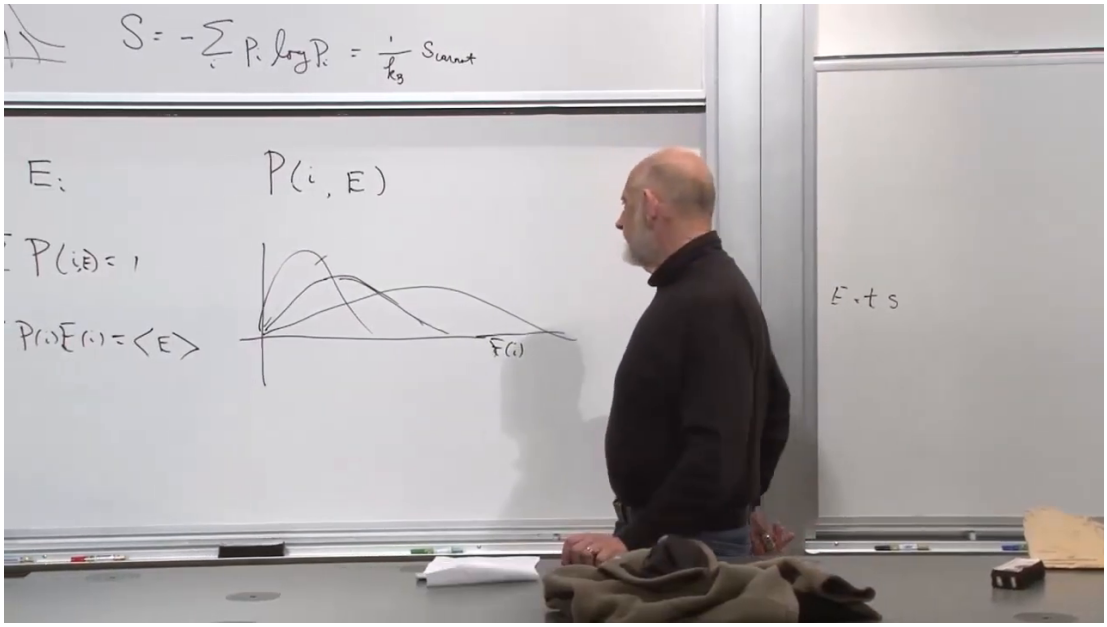


Figure 2.4: Different distributions with different average energies. The right-hand sketch is schematic, but it makes the lecture's point clearly: higher average energy shifts probability weight toward higher-energy states.

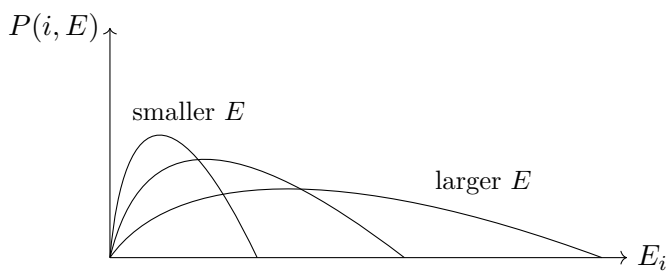


Figure 2.5: A cleaned reconstruction of the family $P(i, E)$. Each curve represents an equilibrium distribution; increasing the average energy shifts the support toward higher-energy states.

That is the role of the next board picture. As the average energy increases, the probability distribution spreads and shifts to the right. The lecture is not yet calculating those distributions. It is using the sketch to make the dependence on average energy visible.

The lecture does not yet prove that higher average energy always means higher entropy. It takes that as a typical feature of ordinary systems: when the distribution spreads over more states, entropy tends to increase. That working assumption will be important in a moment.

2.6 Defining Temperature from the Entropy-Energy Derivative

Now the lecture asks what Susskind himself calls a funny question: how much energy does it take to change the entropy by one bit?

At first the answer sounds almost disappointingly trivial. If energy is a smooth function of entropy,

then

$$dE = \frac{dE}{dS}\, dS. \tag{2.13}$$

Equally well one may write

$$dS = \frac{dS}{dE}\, dE. \tag{2.14}$$

A student objects that perhaps dS/dE is the more natural quantity. Susskind's reply is that either form carries the same information: the two derivatives are inverses of one another when the function is single-valued.

The lecture then promotes the apparently trivial derivative into the central object:

Definition 2.2. Temperature is defined by

$$T = \frac{dE}{dS}. \tag{2.15}$$

Equivalently,

$$dE = T\, dS, \tag{2.16}$$

or

$$dS = \frac{1}{T}\, dE. \tag{2.17}$$

This is the universal definition the lecture wants us to remember. Temperature is a derived quantity. It measures how much energy must be added to produce a small increase in entropy.

The "one bit" question gives a concrete way to read the formula. If the entropy is changed by one bit, that is by $\log 2$, then locally

$$\Delta E \approx T \log 2, \tag{2.18}$$

provided the change is small enough that T may be treated as approximately constant over the step. That is not a new law; it is simply the differential relation read in a practical way.

There is also an important consistency check with the earlier unit conversion. Since

$$T = k_B T_{\text{Kelvin}}, \qquad S = \frac{1}{k_B} S_{\text{Carnot}}, \tag{2.19}$$

the fundamental relation becomes

$$dE = T_{\text{Kelvin}}\, dS_{\text{Carnot}}. \tag{2.20}$$

The two factors of k_B cancel. This is why the same thermodynamic formula appears in ordinary textbooks without an explicit Boltzmann constant.

2.6.1 Question & Answer

Question. Why is it legitimate to regard S as a single-valued function of E? Is that obvious?

Answer. It is not obvious, and the lecture says so explicitly. There are counterexamples. But for normal physical systems it is typically true that increasing the average energy broadens the equilibrium distribution and increases the entropy. For the purposes of this lecture we take that monotonicity as a working assumption. Under that assumption $S(E)$ is single-valued, dS/dE and dE/dS make sense, and the temperature is positive.

This is the point at which the lecture pauses and says, in effect: we now have a definition of temperature, but it is still an abstract one. What does it have to do with the familiar notion of hot and cold? The answer comes from energy flow.

2.7 Two Weakly Coupled Systems and the Direction of Energy Flow

To recover the ordinary meaning of temperature, Susskind considers two systems, A and B, initially each in equilibrium but at different temperatures. They are connected by a thin pipe. The pipe is thin in the thermodynamic sense: the systems can exchange energy, but only weakly, so that each still makes sense as a subsystem with its own entropy and temperature.

Let the initial temperatures satisfy

$$T_B > T_A. \tag{2.21}$$

Which way will the energy flow? The familiar answer is from hot to cold. The lecture now shows that this familiar fact is encoded in the derivative definition of temperature.

Three ingredients are assembled.

First, the total entropy is additive:

$$S_{\text{tot}} = S_A + S_B. \tag{2.22}$$

In the lecture this is justified heuristically by the fact that entropy is the logarithm of a multiplicative quantity.

Second, the total energy is conserved:

$$dE_A = -\,dE_B. \tag{2.23}$$

This is the first law in the present setting.

Third, the second law says that during equilibration the total entropy increases:

$$dS_A + dS_B > 0. \tag{2.24}$$

Now use the definition of temperature separately for each subsystem:

$$dE_A = T_A\,dS_A, \qquad dE_B = T_B\,dS_B. \tag{2.25}$$

Therefore

$$dS_A = \frac{dE_A}{T_A}, \qquad dS_B = \frac{dE_B}{T_B}. \tag{2.26}$$

Substituting into the total entropy change gives

$$\begin{aligned} dS_{\text{tot}} &= dS_A + dS_B && (2.27)\\ &= \frac{dE_A}{T_A} + \frac{dE_B}{T_B} && (2.28)\\ &= \frac{dE_A}{T_A} - \frac{dE_A}{T_B} && (2.29)\\ &= dE_A \left(\frac{1}{T_A} - \frac{1}{T_B} \right). && (2.30) \end{aligned}$$

Now suppose $T_B > T_A$. Then

$$\frac{1}{T_A} - \frac{1}{T_B} > 0. \tag{2.31}$$

Since the second law requires $dS_{\text{tot}} > 0$, it follows that

$$dE_A > 0. \tag{2.32}$$

So system A gains energy while system B loses it. Energy flows from B, the hotter system, to A, the colder one.

If instead $T_A = T_B$, then the coefficient vanishes and there is no preferred direction of flow. That is the equilibrium condition for the composite system.

Remark 2.3. The transcript becomes ragged before the final sign argument is fully written out, but the computation above is the minimal cautious completion of the equations Susskind explicitly sets up: entropy additivity, total-energy conservation, the second law, and $dE = T\,dS$ for each subsystem.

The abstract derivative has now recovered the ordinary physical meaning. Temperature is precisely the quantity whose difference determines the direction of spontaneous energy flow.

2.8 Summary

The lecture begins by refusing to define temperature too quickly. First we are asked to stop thinking in human units. Boltzmann's constant is only a conversion factor: it converts thermometer temperature into energy, and engineering entropy into dimensionless statistical entropy. Once those conversions are made, the formulas simplify and k_B may be suppressed.

Only then does the statistical-mechanical structure emerge. A system in equilibrium is described by probabilities $P(i)$ over microscopic states i, with average energy

$$\sum_i P(i)E_i = E.$$

Equilibrium is then recast as a one-parameter family $P(i, E)$, and each member of that family has its own entropy

$$S(E) = -\sum_i P(i, E) \log P(i, E).$$

From that family comes the central derivative

$$T = \frac{dE}{dS},$$

or $dE = T\,dS$. At first it looks like a trivial identity. Then the lecture reveals that it is the universal definition of temperature. Finally, by coupling two systems weakly and invoking the first and second laws, the lecture shows that this derivative is exactly what governs the flow of energy from hot to cold. The familiar notion of temperature is thus rebuilt from probability, entropy, and energy.

Chapter 3

Entropy, Equilibrium, and the Maximum-Entropy Problem

This lecture begins as a deliberate reset. We go back to the basic objects of statistical mechanics: a discrete set of states, a probability distribution over them, the corresponding energies, and the entropy of that distribution. But the reset is not merely definitional. Very quickly the lecture turns those definitions into a picture of equilibrium, then into the thermodynamic laws, then into a derivation of the direction of heat flow, and finally into the real mathematical problem of the subject: which probability distribution maximizes entropy when the total probability and the average energy are fixed?

Credit. These notes follow Leonard Susskind's lecture, with lecture curation by LazyingArt LLC and Video2Book.

3.1 Entropy, Probability, and Average Energy

We begin exactly as the lecture does. Suppose somebody simply hands us a probability distribution $P(i)$ over a discrete set of states i. We are not yet asking where the distribution came from; that question is postponed. For the moment we only assume that the states are discrete enough that they can be counted.

Each state has a probability $P(i)$, and each state has an energy $E(i)$. The first relation is just normalization,

$$\sum_i P(i) = 1, \tag{3.1}$$

and the second is the definition of the average energy,

$$\sum_i P(i)E(i) = \langle E \rangle. \tag{3.2}$$

On the board this appears in precisely that notation, with $E(i)$ rather than the cleaner E_i. Later, when the algebra becomes heavier, it will be more convenient to write E_i, but at the opening of the lecture it is helpful to keep the board notation in view.

Entropy is then defined by

$$S = -\sum_i P(i) \log P(i). \tag{3.3}$$

The lecture pauses over the minus sign. At first sight the formula looks negative, but it is not. Since $0 < P(i) \leq 1$, each $\log P(i)$ is nonpositive, and therefore every term $-P(i) \log P(i)$ is nonnegative. So the entropy is a positive quantity.

The opening board layout captures all of this at once: the entropy functional at upper left, the two constraints at upper right, and beneath them a family of distributions drawn against energy.

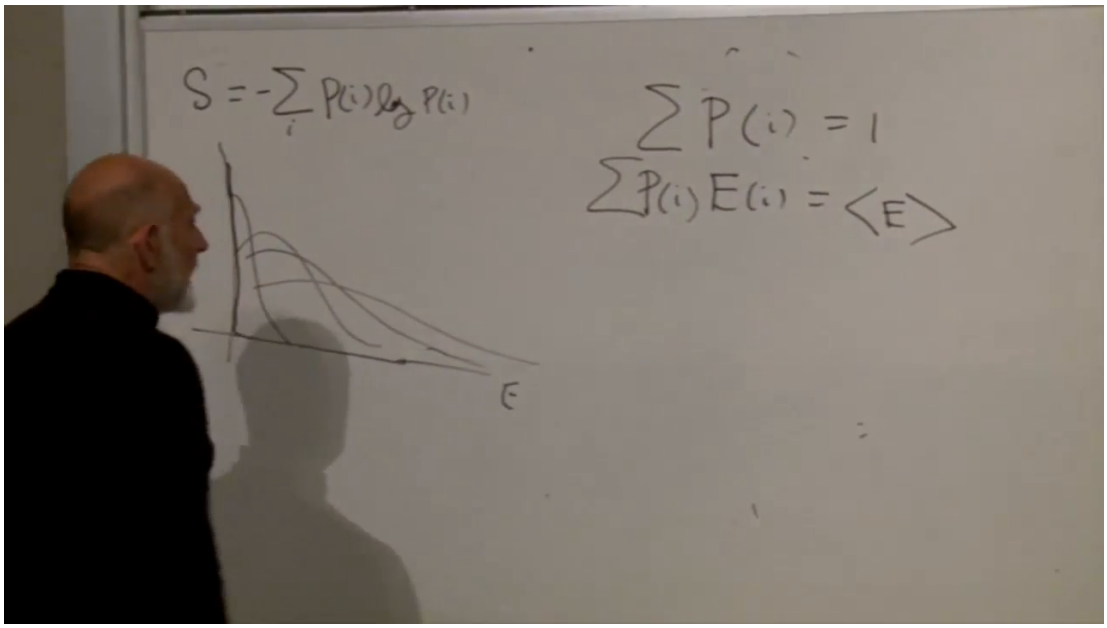


Figure 3.1: Entropy, the normalization and average-energy constraints, and a family of equilibrium probability distributions at increasing average energy.

Already the lecture is telling us that we should not think only of one distribution. We should think of a whole family of them.

3.2 Equilibrium as a Family of Distributions

Susskind immediately recalls the picture from the previous lecture: a one-parameter family of probability distributions, plotted against the energy of the state and parameterized by the average energy. As the distribution broadens and drifts toward larger E, the average energy increases.

There is also a distinguished endpoint of the family. At the minimum possible energy, the probability distribution is concentrated entirely at the ground state. Then there is no uncertainty left about which state the system occupies, and the entropy is exactly zero.

The lecture leans on a qualitative interpretation of entropy here. Entropy is, roughly speaking, a measure of how many states matter underneath the distribution. A very narrow distribution singles out very few important states; a broader one spreads its weight over many of them. Thus, for the equilibrium families we care about, larger average energy comes together with larger entropy.

It is useful to record that as the working assumption

$$\frac{dS}{d\langle E\rangle} > 0, \tag{3.4}$$

but only for the equilibrium family under discussion. This is not a theorem about arbitrary invented distributions.

The board sketch is qualitative, and a clean redraw should stay qualitative as well.

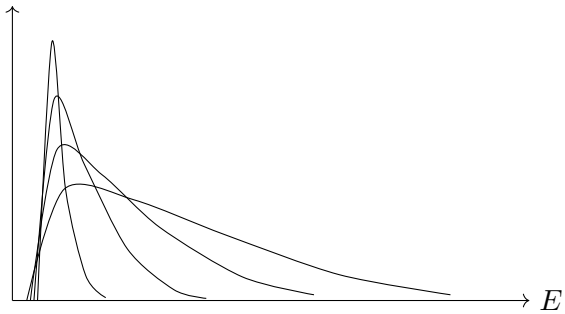


Figure 3.2: Interpretive redraw of the equilibrium family. The board makes the qualitative point: as the average energy increases, the distribution broadens and extends toward larger E.

3.2.1 Question & Answer

Could a distribution move to the right and at the same time get narrower? Certainly. One can invent such a family without difficulty. The lecture says so explicitly. The point is not that probability theory forbids it. The point is that the thermal-equilibrium distributions we are after are not like that. For those distributions, broadening with increasing average energy is exactly the right picture.

3.3 Broadening, Coarse-Graining, and the Thermodynamic Laws

From this picture the lecture pivots to thermodynamics. Thermal equilibrium is characterized by the probability distribution of the states. Of course one may characterize equilibrium by a temperature, but one may also characterize it by an average energy. If a box of gas is in thermal equilibrium, then, in the lecture's phrase, whatever was going to happen has finished happening. Microscopically the molecules still move, but macroscopically the system is quiescent, mixed up, and no longer evolving in any dramatic way.

In that state heat tends not to flow, because all parts of the system are in equilibrium with each other.

Now the laws of thermodynamics can be read in the language of probability distributions. The first law is simply energy conservation. The second law says that entropy increases. But what does that mean in the present setting? It means that when a system starts away from equilibrium and is allowed to come to equilibrium, the probability distribution generally broadens.

This is where the lecture introduces coarse-graining. To coarse-grain is, in effect, to be a little careless about the exact microscopic information. Once we lose track of details, the distribution tends to spread rather than sharpen. A narrowing distribution would mean that somehow we had

learned more and more about an enormous complicated system without actually measuring it. That is not the natural direction of things. The natural direction is that we lose track, the distribution broadens, and entropy grows.

The zeroth law is then presented in a modernized way. Historically it entered thermodynamics axiomatically, but here the lecture treats it as illuminated by the first and second laws together with the definition of temperature already introduced. There exists a notion of temperature such that energy flows from higher temperature to lower temperature, and in equilibrium all parts of the system have the same temperature.

The lecture now asks the natural question: does the earlier definition of temperature really know which way heat flows? Let us test it.

3.4 Temperature and the Direction of Heat Flow

Take a system made of two connected parts, A and B, able to exchange energy. Assume, just to fix conventions, that

$$T_B > T_A. \tag{3.5}$$

If the opposite inequality held, we would merely interchange the labels.

The first law gives

$$dE_A + dE_B = 0, \tag{3.6}$$

because the total energy of the combined system is conserved.

The second law gives

$$dS_A + dS_B \geq 0, \tag{3.7}$$

with equality only in special cases. Generically the total entropy increases.

Now use the earlier definition of temperature,

$$dE = T\,dS. \tag{3.8}$$

Applied separately to the two subsystems,

$$dE_A = T_A\,dS_A, \qquad dE_B = T_B\,dS_B. \tag{3.9}$$

Substituting these into energy conservation, we obtain

$$T_A\,dS_A + T_B\,dS_B = 0. \tag{3.10}$$

Solve for dS_B:

$$dS_B = -\frac{T_A}{T_B}\,dS_A. \tag{3.11}$$

Now eliminate dS_B from the entropy inequality:

$$dS_A + dS_B \geq 0, \tag{3.12}$$

$$dS_A - \frac{T_A}{T_B} dS_A \geq 0, \tag{3.13}$$

$$\left(1 - \frac{T_A}{T_B}\right) dS_A \geq 0. \tag{3.14}$$

For the moment we assume temperatures are positive. Multiplying by $T_B > 0$ gives

$$(T_B - T_A)dS_A \geq 0. \tag{3.15}$$

But $T_B - T_A > 0$ by assumption. Therefore

$$dS_A > 0. \tag{3.16}$$

Multiplying once more by T_A, we get

$$dE_A = T_A\, dS_A > 0. \tag{3.17}$$

So subsystem A gains energy, and therefore subsystem B loses it. Energy flows from B to A, that is, from the hotter subsystem to the colder one.

This is the point of the calculation. Starting from energy conservation, entropy increase, and the definition $dE = T\, dS$, we recover the physical meaning of temperature: it determines the direction of heat flow. In the lecture, when Susskind says "heat" here, he simply means energy transfer.

The flow continues until the temperatures become equal. When

$$T_A = T_B, \tag{3.18}$$

there is no preferred direction for further transfer, and thermal equilibrium is established. The transitive part of the zeroth law then becomes obvious: if A is in equilibrium with B, and B with C, then all three have the same temperature.

3.4.1 Question & Answer

Does this derivation assume that A and B are each individually in equilibrium? Yes, it does. The very use of T_A and T_B presupposes that each subsystem is internally equilibrated enough for a temperature to be assigned to it. If that is not true, then one must subdivide further. The same reasoning is then applied to smaller pieces until the local equilibrium assumption becomes appropriate.

3.5 What a State Means

Only after the heat-flow argument is finished does the lecture stop and ask a more basic question: what is a state?

A state, we are told, means as much as one can possibly know about the system if one were infinitely powerful, subject always to the actual laws of physics. That last clause matters. In classical mechanics one is tempted to say that a state of a gas is the complete list of positions and momenta of all the molecules. As a first picture, that is the right thing to have in mind. But quantum mechanics places limits on what can be known simultaneously, and in the quantum description the state counting for a finite box becomes discrete.

So there are really two layers here. The conceptual layer is invariant: a state is the most refined description allowed by the theory. The technical layer depends on context: in a quantum treatment we sum over discrete states, while in a classical limit those sums are replaced by integrals. The lecture emphasizes that passing carefully from the quantum description to the classical one is subtle, but the broad statement is correct.

3.5.1 Question & Answer

Should we literally think of the state of a gas as the position and momentum of every molecule? In the classical picture, yes, that is the right image. But one should immediately add the lecture's caveat: quantum mechanics changes what "as much as can be known" means. The useful definition is therefore the more general one. A state is not simply a list of classical variables; it is the finest specification of the system permitted by the laws of physics.

3.6 Replicas, Occupation Numbers, and Maximum Entropy

Now we come to what the lecture itself calls the hard question. Once equilibrium has been reached, what is the probability distribution $P(i)$ over the microstates?

To get a handle on that question, it is useful to imagine that the system of interest is not isolated but in contact with a very large reservoir, a heat bath, with which it can exchange energy. Because the bath is enormous, a small flow of energy into or out of the subsystem barely changes the bath's temperature.

A particularly convenient model of the heat bath is then introduced. Instead of an amorphous reservoir, imagine a very large number N of identical copies of the subsystem, all weakly connected so that energy can flow among them. One of them is the subsystem we care about; the rest collectively play the role of the bath. This is not merely a trick. In many physical situations a large system really can be regarded as built from many similar smaller subsystems.

From Counting to Probability. Each replica is in some state i. Let n_i denote the occupation number, the number of replicas found in the i-th state. The board at this point shows an energy axis with many discrete level marks and writes the basic identification very clearly:

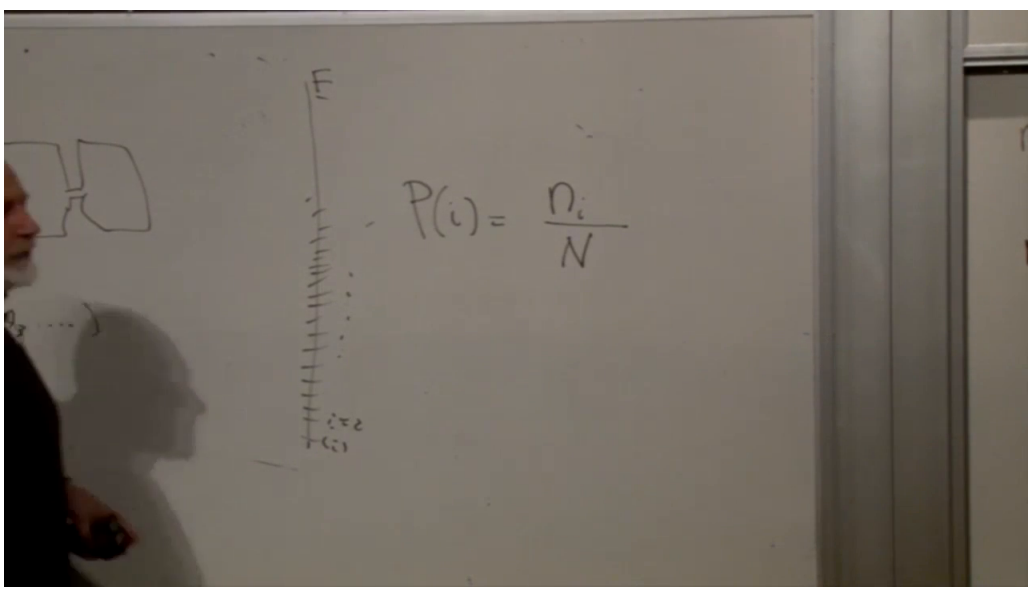


Figure 3.3: Probability as the occupation fraction $P(i) = n_i/N$.

Thus

$$P(i) = \frac{n_i}{N}. \tag{3.19}$$

In words: the probability that a given subsystem is in state i is simply the fraction of the replicas that are in that state.

From here on it is convenient to write the state energies as E_i. The occupation numbers satisfy two constraints. First, the total number of replicas is fixed:

$$\sum_i n_i = N. \tag{3.20}$$

Second, the total energy is fixed. If E denotes the average energy per subsystem, then

$$\sum_i n_i E_i = NE. \tag{3.21}$$

Using $n_i = NP_i$, these become

$$\sum_i P_i = 1, \qquad \sum_i P_i E_i = E. \tag{3.22}$$

So the familiar constraints on probabilities are just the occupation-number constraints rewritten per subsystem.

Multiplicity. The next question is purely combinatorial: for a given set of occupation numbers $\{n_i\}$, how many ways are there to distribute the N replicas among the states so as to realize exactly that set?

The answer is

$$C = \frac{N!}{\prod_i n_i!}, \tag{3.23}$$

and the lecture suggests calling this C, for the combinatorial factor. If some occupation numbers are zero, nothing blows up, because

$$0! = 1. \tag{3.24}$$

So unoccupied states simply contribute a factor of one to the denominator.

The lecture pauses long enough to check the formula against simple cases. If all N replicas lie in one state and all other occupation numbers vanish, then

$$C = \frac{N!}{N!} = 1, \tag{3.25}$$

as it should: there is only one way to put everything in one cup. If three replicas occupy three different states, then

$$C = \frac{3!}{1!\,1!\,1!} = 6. \tag{3.26}$$

Again that is exactly right.

Why do we care about C? Because of a symmetry argument. If all microscopic redistributions compatible with the setup are treated symmetrically, then the most probable macroscopic occupation pattern is the one realized in the largest number of ways. So the statistical-mechanical problem becomes: maximize C, but maximize it subject to the two constraints above.

Stirling's approximation. To do that we need an approximation for large factorials. The lecture takes time to derive the simplified form of Stirling's approximation that is sufficient for this problem.

Start from

$$\log N! = \sum_{x=1}^{N} \log x. \tag{3.27}$$

For large N, replace the sum by an integral:

$$\log N! \approx \int_1^N \log x \, dx \tag{3.28}$$

$$= [x \log x - x]_1^N \tag{3.29}$$

$$= N \log N - N + 1. \tag{3.30}$$

The additive constant is negligible on the scale of N, and the lecture keeps only the leading terms:

$$\log N! \approx N \log N - N. \tag{3.31}$$

Exponentiating,

$$N! \approx N^N e^{-N}. \tag{3.32}$$

The familiar $\sqrt{2\pi N}$ factor, together with other subleading corrections, is deliberately omitted. For the present argument only the leading exponential behavior matters.

One may ask whether Stirling's approximation is justified in the denominator as well as the numerator. The lecture's answer is that, in the large-N regime of interest, the relevant occupation numbers scale proportionally with N. As the total number of replicas grows, the typical n_i grow with it, so the same approximation applies to the important denominator terms.

From multiplicity to entropy. Apply the approximation upstairs and downstairs:

$$C \approx \frac{N^N e^{-N}}{\prod_i n_i^{n_i} e^{-n_i}}. \tag{3.33}$$

Because $\sum_i n_i = N$, the exponential factors cancel, leaving

$$C \approx \frac{N^N}{\prod_i n_i^{n_i}}. \tag{3.34}$$

Now take the logarithm:

$$\log C = N \log N - \sum_i n_i \log n_i. \tag{3.35}$$

Substitute $n_i = NP_i$:

$$\log C = N \log N - \sum_i NP_i \log(NP_i) \tag{3.36}$$

$$= N \log N - N \sum_i P_i (\log N + \log P_i) \tag{3.37}$$

$$= N \log N - N \log N \sum_i P_i - N \sum_i P_i \log P_i. \tag{3.38}$$

Using $\sum_i P_i = 1$, the $N \log N$ terms cancel, and we find

$$\log C = -N \sum_i P_i \log P_i = NS. \tag{3.39}$$

This is the decisive turn of the lecture. Maximizing the number of ways of rearranging the replicas is the same as maximizing the entropy. Therefore the most probable occupation-number distribution is obtained by solving

$$\text{maximize } S(P_i) = -\sum_i P_i \log P_i \tag{3.40}$$

subject to

$$\sum_i P_i = 1, \qquad \sum_i P_i E_i = E. \tag{3.41}$$

That is the statistical-mechanical rule.

3.6.1 Question & Answer

By symmetry, shouldn't all the occupation numbers simply be equal? They would, if there were no energy constraint. Without constraints, the arithmetic does push toward uniformity. But the states have different energies, and the condition

$$\sum_i P_i E_i = E \tag{3.42}$$

breaks that symmetry. The asymmetry does not come from nowhere; it comes from the fact that not all states cost the same energy.

3.7 Lagrange Multipliers and the Problem Left for Next Time

The lecture does not solve the entropy-maximization problem yet. Instead it spends the rest of the time introducing the tool needed to solve it: Lagrange multipliers.

Without constraints, the rule for extremizing a function $F(x_1, x_2, \dots)$ is familiar:

$$\frac{\partial F}{\partial x_i} = 0 \qquad \text{for all } i. \tag{3.43}$$

But here the extremization is constrained. One wants to maximize or minimize a function while demanding that some other function vanish:

$$G(x_1, x_2, \dots) = 0. \tag{3.44}$$

The lecture first explains the geometry. In two variables, the level curves of F are like contours on a map, while $G = 0$ is a constraint curve cutting across them. The constrained extremum lies where a level curve of F is tangent to the constraint curve.

The analytic trick is the devilishly clever part. Instead of extremizing F subject to $G = 0$, define

$$F' = F + \lambda G, \tag{3.45}$$

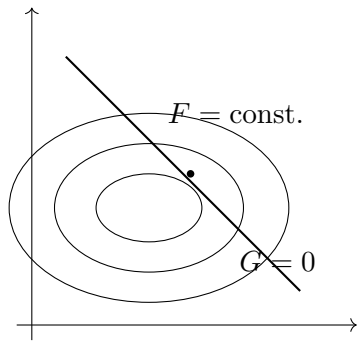


Figure 3.4: Interpretive contour sketch for one constrained extremization problem. The constrained extremum occurs at tangency.

where λ is a new unknown, the Lagrange multiplier. One then extremizes F' as though there were no constraint, and afterwards imposes $G = 0$ to determine λ.

The lecture works through a simple example:

$$F = \frac{x^2 + y^2}{2}, \qquad G = x + y - 1. \tag{3.46}$$

So

$$F' = \frac{x^2 + y^2}{2} + \lambda(x + y - 1). \tag{3.47}$$

Now set the derivatives of F' to zero:

$$\frac{\partial F'}{\partial x} = x + \lambda = 0, \qquad \frac{\partial F'}{\partial y} = y + \lambda = 0. \tag{3.48}$$

Hence

$$x = y = -\lambda. \tag{3.49}$$

The remaining equation is simply the original constraint,

$$x + y = 1. \tag{3.50}$$

Therefore

$$-2\lambda = 1, \qquad \lambda = -\frac{1}{2}, \tag{3.51}$$

and so

$$x = y = \frac{1}{2}. \tag{3.52}$$

That is the point on the line $x + y = 1$ where $(x^2 + y^2)/2$ is minimized.

With several constraints one proceeds in the same way:

$$F' = F + \lambda_1 G_1 + \lambda_2 G_2 + \cdots, \tag{3.53}$$

and then solves the stationarity equations together with

$$G_1 = 0, \qquad G_2 = 0, \qquad \ldots \tag{3.54}$$

Now the real statistical-mechanical problem can be written in precisely that language. We want to maximize

$$S(\{P_i\}) = -\sum_i P_i \log P_i \tag{3.55}$$

subject to the two constraints

$$G_1 = \sum_i P_i - 1 = 0, \qquad G_2 = \sum_i P_i E_i - E = 0. \tag{3.56}$$

That is the mathematics problem the lecture leaves for next time. Once those equations are solved, the equilibrium probability distribution is known.

The lecture also makes a broader point here. At this stage we are not really doing the theory of gases or superconductors in any detailed sense. We are doing probability theory: maximizing an entropy, or equivalently a multiplicity, subject to constraints. The same mathematics appears in many contexts. Statistical mechanics is the physical arena in which those constrained probability problems become unavoidable.

And there is one more hint before the lecture ends: temperature itself will turn out to be a Lagrange multiplier.

3.8 Summary

What began as a recap has turned into a precise program. We started from a probability distribution over states, wrote its average energy and entropy, and pictured a family of equilibrium distributions whose entropy grows with energy. We then translated the thermodynamic laws into the language of distributions and used $dE = T\,dS$ to show that energy flows from hotter to colder subsystems until their temperatures equalize. After that the lecture changed point of view: from one system to many replicas, from probabilities to occupation numbers, and from occupation numbers to multiplicities. The key result was

$$\log C = NS, \tag{3.57}$$

so that the most probable distribution is the one that maximizes entropy subject to normalization and fixed average energy. The lecture stops there, with the right problem fully posed and the method of Lagrange multipliers ready for the next step.